Hagar Soliman Hamdan

AI Engineer

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LinkedIn | Portfolio | GitHub | Upwork

Summary

AI Engineer with a Bachelor's degree in Artificial Intelligence and hands-on experience building machine learning and deep learning solutions across NLP, Computer Vision, and Generative AI. Experienced in developing, fine-tuning, and evaluating AI models and applications using Python, PyTorch, TensorFlow/Keras, Transformers, OpenCV, and Hugging Face. Built practical AI solutions including sentiment analysis, multilingual review analysis, eye tracking, real-time object detection, and LLM-based applications, with a focus on turning AI models into functional, real-world solutions.

Experience

AI Integration — CRM 07/2026 – Present | Remote | Cairo, Egypt
Freelance Client

- Integrated and customized an OpenAI-powered AI chatbot within Perfex CRM to automate customer interactions and AI-based responses.
- Implemented human handoff workflows with staff routing, permissions, and live support integration for seamless AI-to-human conversations.
- Configured Pusher, SMTP, and CRM communication workflows to support real-time messaging, notifications, and email operations.
- Configured WebRTC voice/video calling with TURN server integration to enable reliable cross-network communication.

AI Engineer — Freelance 04/2026 – 05/2026 | Remote | Cairo, Egypt
Upwork

- Developed sentiment analysis models using Python, TensorFlow/Keras, RNN, and BiLSTM.
- Applied NLP preprocessing and model evaluation for automated sentiment classification.
- Delivered the project successfully with a **5.0/5.0 client rating**.

Data Labeling 05/2024 – 09/2024 | Cairo, Egypt
MEEG Company

- Annotated **3,000+ road-scene images** for Computer Vision datasets.
- Created accurate bounding-box annotations for object detection tasks.
- Prepared high-quality labeled data for AI model training and evaluation.

Education

Bachelor of Artificial Intelligence 10/2022 – 07/2026 | Cairo, Egypt
Helwan International Technological University
Overall Grade: 90.40% — Excellent, Ranked **8th** in the graduating class

Technical Skills

Programming Languages
Python, PHP, JavaScript, HTML, CSS

Backend & APIs
FastAPI, Flask, Streamlit

AI & Development Tools
ChatGPT, Claude Code, Antigravity, Visual Studio Code, Git, GitHub, Docker, Jupyter, Google Colab,XAMPP

Artificial Intelligence
Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, LLMs, Transformers, Hugging Face, RAG, Embeddings, Sentiment Analysis, Text Classification, Object Detection, Image Processing,Prompt Engineering, Eye Tracking, Scikit-learn, TensorFlow/Keras, PyTorch, OpenCV, YOLO

Databases
MySQL, SQL

Projects

EyeSpeak — AI Eye Tracking Communication System
Graduation Project
Developed a hands-free interaction system using Computer Vision and Machine Learning for eye-controlled communication.
Tools: Python, OpenCV, Random Forest, Raspberry Pi Zero 2 W, Android(Kotlin), IoT
Result: 87% eye-tracking accuracy

Smart Product Review Analyzer
NTI Training Project
Built a multilingual system to analyze Arabic and English Amazon reviews, classify sentiment, and rank products.
Tools: Python, PyTorch, Transformers, Hugging Face, Streamlit
Result: 92% accuracy, 0.89 F1-score

Vision Walk — Real-Time Pedestrian Detection
Shabab Mobtakeron Training Project
Built a real-time Computer Vision system for pedestrian detection and tracking in road scenes.
Tools: Python, YOLO, OpenCV
Result: ~80% detection accuracy

AI-Powered Sign Language Translator Smart Glove
Developed a smart glove using Machine Learning to recognize hand gestures and translate them into Arabic text and speech.
Tools: Python, Random Forest, ESP32, Flutter
Result: 87% gesture recognition accuracy

Certificates

Digital Egypt Pioneers Initiative (DEPI)
Generative AI Track

NVIDIA
AI for All: From Basics to GenAI Practice

ITI MaharaTech
Database Fundamentals

National Telecommunication Institute (NTI)
Artificial Intelligence Training Program

IBM
Machine Learning with Python

ITI MaharaTech
Applied Deep Learning

NVIDIA
Building LLM Applications With Prompt Engineering

Shabab Mobtakeron - Ministry of Youth and Sports, Egypt
Introduction to Artificial Intelligence & Machine Learning with Quantum Computing Foundations

Languages

- Arabic
- English